CUSTOMER SUPPORT COPILOT

RAG Architecture Comparison and Evaluation Report

Dandes Electronics Knowledge Base

Evaluation date: 25 October 2026

Evaluation set: 20 questions across three RAG versions

This report summarizes the recorded retrieval metrics, custom LLM-as-Judge results, and RAGAS scores generated by the project evaluation scripts.

1. Executive Summary

The project compares V1 Basic RAG, V2 Single-Index RAG, and V2 Multi-Index RAG using the same 20-question test set. The multi-index approach achieved the highest Recall@5 (94.38%) and Routing Accuracy (95.00%), with zero false refusals. Its RAGAS faithfulness score (0.8062), however, is below the target of 0.85, so answer grounding remains an area for improvement.

The results show trade-offs rather than a universal winner: V2 Single-Index has the highest Precision@5 and multi-source Coverage, while V2 Multi-Index has the strongest Recall@5, routing, Context Precision, and Context Recall.

2. System Versions

V1 Basic RAG. Baseline retrieval and answer generation using the shared knowledge base.

V2 Single-Index RAG. Advanced retrieval techniques including metadata filtering, hybrid search, reranking, and query transformation.

V2 Multi-Index RAG. Routes questions to specialized indexes for policy, product, order, and resolved-ticket information. Out-of-scope questions should receive the configured refusal response.

3. Retrieval and Refusal Metrics

Metric	V1 Basic RAG	V2 Single-Index	V2 Multi-Index
Precision@5	28.75%	30.00%	28.75%
Recall@5	91.25%	92.50%	94.38%
Coverage (multi-source)	94.29%	97.14%	87.14%
Routing Accuracy	N/A	N/A	95.00%
Refusal Accuracy	100.00%	100.00%	100.00%
False Refusals	1	1	0
Questions Evaluated	20	20	20

Interpretation. Multi-index routing reached 95% accuracy and avoided false refusals in this test set. Its multi-source coverage was 87.14%, below both other versions, suggesting that source selection or aggregation needs attention.

4. Generation Evaluation

The project uses a custom LLM-as-Judge evaluation and RAGAS. The custom judge's strict label-based faithfulness rate is not numerically equivalent to the RAGAS faithfulness score.

Metric	V1 Basic RAG	V2 Single-Index	V2 Multi-Index
LLM-Judge Faithfulness	93.75%	75.00%	37.50%
Answer Correctness (LLM Judge)	81.25%	68.75%	43.75%
RAGAS Faithfulness	0.8786	0.8832	0.8062
Answer Relevancy (RAGAS)	0.7270	0.7126	0.7129
Context Precision (RAGAS)	0.8410	0.9156	0.9448
Context Recall (RAGAS)	0.8646	0.8646	0.9146
In-scope questions (RAGAS)	16	16	16

Interpretation. V2 Multi-Index has the strongest RAGAS Context Precision (0.9448) and Context Recall (0.9146), indicating comparatively targeted retrieval and better coverage of reference information. Its RAGAS faithfulness is 0.8062, below the 0.85 target. The custom judge also flags lower faithfulness and correctness for this version; inspect answers against their retrieved contexts.

5. Target Assessment for V2 Multi-Index

Target	Recorded result	Assessment
Routing Accuracy >= 85%	95.00%	Met
Recall@5 >= 0.80	94.38%	Met
RAGAS Faithfulness >= 0.85	0.8062	Not met
Refusal Accuracy = 100%	100.00%	Met
False Refusals = 0	0	Met

6. Limitations and Recommended Improvements

• Improve grounding: require answers to use retrieved evidence, cite source IDs, and abstain when evidence is insufficient.

• Investigate judge failures: compare low-scoring answers against the retrieved passages and reference answers.

• Improve multi-source coverage: verify that the router selects every needed index and that relevant results survive ranking and context assembly.

• Confirm Q01-Q07 match the assignment's Section 7 questions, reference answers, source IDs, and expected indexes.

• Report custom LLM-as-Judge label rates separately from RAGAS scalar scores because their definitions differ.

7. Reproducibility and Artifacts

The evaluation directory contains the golden test set and evaluation scripts. Per-version result files and the retrieval, generation, and RAGAS metrics are stored in evaluation/results/. The comparison table is generated at evaluation/comparison_table.csv. The demo output belongs in the project root as demo_output.txt.

Keep the API key in the local root .env file. Do not include .env in a public repository or submission archive. Follow README.md to rerun the project and evaluation scripts.

Conclusion

The recorded results support V2 Multi-Index RAG for routing and retrieval recall, but answer grounding remains unresolved. It meets four of the five stated multi-index targets; RAGAS faithfulness must improve before all evaluation goals are met.